



LITERATURE REVIEW AND COMPETITOR ANALYSIS

MScFE Capstone Project

Opolot Simon Peter (opolots500@gmail.com)

Uchenna Luke (luke.ezima@gmail.com)

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Problem Statement:

1. PROBLEM STATEMENT

Every investor who has already raised capital faces the same practical question: how should the money be split across assets so that it survives shocks and still grow? Brinson, Hood and Beebower(1986) showed that the long -run allocation policy explains most of the ups and downs in a pension fund's returns overtime, and Ibbotson and Kaplan (2000) confirmed this, while noting that differences between funds depend more on how each fund carries its policy out. Either way, the split matters.

The standard tools for making that split have known weaknesses. Mean-variance optimization (Markowitz,1952) reacts violently to small errors in its inputs. Simply splitting money equally across assets (the "1/N" rule) is surprisingly hard to beat (DeMiguel, Garlappi & uppal,2009). Hierarchical Risk Parity (HRP) avoids return forecasts and group assets by how they move together (Lopez de Prado, 2016), but it is a fixed rule that only looks backwards over one historical window. These weaknesses showed up in practice: in 2022 stocks and bonds fell together, and the classic 60/40 portfolio lost on both sides at once.

Deep learning promises a model that can learn when shift weights as conditions change (Zhang,Zhohren &Roberts,2020). But the published evidence is thin in exactly the places that matter to a real investor: short test periods, trading costs ignored or set very low, and comparisons against weak rules rather than HRP. Our study as a plain question: after costs and out of sample, does a deep learning allocator really do better that the best simple rule, or does it only look better because of how it was tested.

We train Long Short-Term Memory (LSTM), network, a neural network designed for sequences of data, to output weekly, long-only weights for eight liquid exchange funds (ETF's), with no asset above 40%. It is trained to maximize the Sharpe ratio (return earned per unit of risk) of its weekly holding-period returns after trading costs, with turnover measured at the weekly rebalancing dates, so that training matches the way the strategy trades. It is tested year by year for 2015 to 2025, always using only date available at the time.

Our design is closest to Zhang, Zohren & Roberts, (2020), who train an LSTM that outputs weights directly with a Sharpe-ratio loss on a small multi-asset universe. Relative to that study, we add three things: HRP as the primary benchmark, the LSTM must beat (RQ1); macro-financial inputs, whose value we test against a price -only model (RQ2); and a longer, cost-aware out-of-sample test form 2015 tp 2015 that includes 2020 and 2022, with cost built into the training loss at the weekly horizon at which the strategy trades and stressed from 0 to 50 basis points(RQ3).

Hypothesis and research questions

Item	Statement
H0	Net of a10 basis-point (0.10%) trading cost, the LSTM allocator's out-of-sample Sharpe ratio is higher than HRP's.
H1	Net of q 10 basis point trading cost, the LSTM allocator's out-of-sample Sharpe ratio exceeds HRP's.
RQ1	Does the LSTM beat HRP, 1/N,60/40. Minimum-variance and maximum Sharpe allocation on a risk-adjusted basis, a cost -adjusted basis, or both?
RQ2	Do macro financial inputs (market volatility, the yield curve, the US dollar) add value beyond price history alone?

RQ3	Do the results hold across sub periods (2015-2019,2020,2021,2022 and 2023 -2025), stress episodes, cost levels and asset sets?
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Table 1. Study design (unchanged from the problem statement except were marked)

Element	Specification
Assets	SPY (US stocks), EFA (developed ex-US stocks), EEM (emerging -markets stocks), TLT (long Us treasuries), LQD(corporate bonds), GLD (gold), DBC (commodities), VNQ (US real estate)
Data	Daily adjusted prices, jan 2007- Dec 2025 (Yahoo Finance); VIX, 10y 2y Treasury spread and broad dollar index from FRED, lagged one day
Model	LSTM, one year,32 or 64 units, dropout 0.2, 60-day look -back; trained on weekly holding-period returns with a Sharpe loss net of cost (annualized over 52 weeks) plus a penalty on turnover measured at the weekly rebalancing dates; weights averaged over 10 random seeds
Benchmarks	HRP (primary), 1/N,60/40, minimum-variance and maximum -Sharpe with Ledoit -Wolf covariance (252-day window); + 63- day inverse-volatility rue as a cheap adaptivity check
Trading	Decide at Friday close, trade at the next day 's close; 10 bps per unit traded, charged in the same wat in training and in the backtest; 40% cap for all rules except fixed 1/N and 60/40
Testing	Train 2007 -13, validate 2014, then yearly out-of-sample folds 2015- 2025, retaining each January; 5- day purge between training, validation and test data
Sub-periods	2015 – 2019,2020,2021,2022 and 2023-2025, each reported separately; 2008 lies inside the initial training window and is not tested out of sample
Judging	Net return, volatility, Sharpe, maximum drawdown, Calmar, 95% CvaR, turnover; Ledoit-Wolf (2008) Sharpe test; Deflated Sharpe ratio; + break-even cost; + stress-window scorecard

2. Literature Review

2.1 Introduction

This research asks what the research to date tells us about one practical decision: Is extra model complexity worth paying for when allocating across asset classes? We grouped the work into five themes, moving from the classic estimation problem, through rules that use structure instead of forecasts, to models that learn weights directly, and finally to how such models should be tested and the market shift of 2022 that puts them under pressure. For each theme, we ask what was shown, how convincing it is, and what it leaves open for us. A small group of researchers shaped this field. Harry Markowitz set the framework; Richard Michaud; Victor DeMiguel, Lorenzo Garlappi and Raman Uppal showed how badly it can behave I practice; Olivier Ledoit & Michael Wolf supplied the tools to repair its inputs and to test Sharpe ratios properly; Marcos lopez ded Prado & Daviod Bailey supplied HRP and the main test against backtest overfitting; and Bryan Kelly, Dacheng Xiu, Stefan Zohrfen & Stephen Roberts lead the machine-learning work closet to ours.

2.2 Theme 1 -The estimation problem: Why optimizers disappoint

Markowitz (1952) framed allocation as choosing the mix with the highest expected return for a given risk. The logic is sound, but it needs expected returns and covariance matrix (how each pair of assets moves together) and both must be estimated from noisy history. Michaud (1989) called optimizers “error maximizers”, they put the most money exactly where the estimation errors are most flattering. Ledoit & Wolf (2004) addressed half of the problem with shrinkage, pulling the noisy covariance estimate towards a simpler more stable one. This clearly helps but it does nothing about the harder half, the expected return forecasts.

DeMiguel et al, (2009) then compared 14 optimized models with $1/N$ on seven datasets and found that none consistently beat $1/N$ out of sample. They estimated that, for 25 datasets, about 3,000 months of data would be needed before optimization reliably wins. Kritzman, Page & Turkington, (2010) pushed back, showing that optimization does beat $1/N$ when inputs are set sensibly (for example long run rather than short rolling estimates of return). Read together, the two papers teach something more useful than “ $1/N$ wins”: the verdict depends heavily on the test design. Short windows, noisy inputs and loose testing can make any method look good or bad. For our study, this means two things, the benchmark set must include both the equal -weight rule and a properly estimated optimizer, and the testing rules must be fixed before we see any results.

2.3 Theme 2- Using structure instead of forecasts

A second line of work avoids return forecasts altogether. Lopez de Prado (2016) proposed: group assets by how closely they move together, then split the risk step by step down the groups. Because it never inverts the full covariance matrix, it stays stable even when estimates are noisy, and in his tests, it produced lower out-of-sample risk than classical optimizers. Raffinot (2017) found that portfolios built on similar clustering ideas compare well with traditional risk-based methods out of sample. Risk parity more generally, which gives each asset a similar share of total risk, is now a mainstream institutional strategy (Asness, Frazzini & Pedersen, (2012). This is why HRP, is not $1/N$, is our primary benchmark: it is the strongest simple rule a practitioner could adopt today.

HRP has one clear limit: it looks only at the past year and reacts slowly. But the literature also shows that reacting to changing conditions does not have to be complicated. Moreira & Muir, (2017) found that simply holding less when volatility is high improve the Sharpe ratios of many equity strategies. Cedeberg, O’Doherty, Wang & Yan, (2020) later showed these gains are much less reliable once the tests are made fully out of sample. The lesson for us is important and often ignored: much of the “adaptivity” a neural network offers may be nothing more than reacting to recent volatility, something a three-line rule can do. A deep-learning allocator should therefore be asked to beat a cheap volatility-aware rule, not only a fixed one.

2.4 Theme 3- Learning the weights directly

The idea of training a model directly on a trading objective is older than deep learning, Moody & Saffell, (2001) trained trading systems to maximize a Sharpe-type measure instead of forecasting prices, arguing that forecast accuracy and trading profit are not the same thing. LSTM networks (Hochreiter & Schmidhuber, 1997) later made it practical to learn from long sequences. Fischer & Krauss, (2018) applied LSTMs to S&P 500 stock and beat other machine-learning models, but their profits largely faded after 2010 once costs were counted, an early warning that an edge found in one period may not last. Gu, Kelly and Xiu, (2020) showed that neural networks predict stock returns better than linear models because they capture interactions between signals, yet the predictable part of monthly returns is below one percent, the signals exist, but it is small, and small signals are easily eaten by trading costs.

Zhang, Zohren & Roberts, (2020) is the study closest to ours. They trained a single-layer 64-unit LSTM to output weights for four instruments (VTI, AGG, DBC and the VIX index) by maximizing the Sharpe ratio, and reported higher Sharpe ratios than fixed mixes and mean-variance portfolio over 2011 to

April 2020. It is clean and influential design, and adopt its core idea. On close reading, however, four points limit what it can tell an investor.

- One of the four assets, the VIX index, cannot be bought directly. Holding volatility requires futures or exchange-traded products that carry large roll costs, so part of the results is not investable, our eight assets are all ordinary, liquid ETFs.
- A volatility- targeting overlay (10% target) was applied to the strategies. This makes it hard to tell how much of the gain came from the network and how much from the volatility scaling, which is exactly the concern raised in theme 2.
- The benchmarks did not include HRP or any other modern risk-based rule, and costs were tested at 1 and 10 basis points only.
- The test ends in April 2020, before the 2022, before the 2022 regime in which stocks and bonds fell together.

Our study therefore departs from Zhang, Zohren & Roberts, (2020) in three ways, which together form our contribution. First HRP, rather than fixed mixes and mean variance portfolios, is the benchmark the LSTM must beat (RQ1). Second, we add macro-financial inputs (market volatility, the yield curve and the US dollar) and test their value against a price-only model (RQ2). Third, the out-of-sample test is longer and cost-aware: it runs from 2015 to 2025 through 2020 and 2022, with costs built into the training loss at the weekly horizon at which the strategy actually trades, and stressed from 0 to 50 basis points (RQ3).

Other work pushes the frontiers in different directions. Jiang, Xu & Liang, (2017) used deep reinforcement learning, where the model learns by trial and error, to manage a portfolio of cryptocurrencies at 30 intervals; the market and timescale are far from those of a multi-asset investor. Uysal, Li & Mulvey, (2024) combined a neural network with risk-budgeting step and reported an out-of-sample Sharpe ratio of 1.16 over 2017- 2021, against 0.79 for risk parity and 0.83 for equal weights. This hybrid “structure plus learning” approach is the current cutting edge, but again the test window is five years and does not contain 2022.

2.5 Theme 4 Testing honesty

With flexible models, it is easy to fool ourselves., if enough settings are tried, one will look good by chance. Bailey & Lopez de Prado, (2014) proposed the deflated Sharpe ratio, which lowers a reported Sharpe ratio according to how many trials were run and how unusual the returns are. Bailey, Borwein, Lopez de Prado & Zhu, (2017) showed how to estimate the probability that a backtest is overfitted, and Harvey, Liu & Zhu, (2016) argued that, because so many strategies have been tested, a new finding should clear a t-statistic of about 3.0 rather than the usual 2.0. To compare two Sharpe ratios fairly when returns are not normally distributed, Ledoit & Wolf, (2008) provide robust test.

These tools are well known, yet they are rarely used in deep-learning allocation papers reviewed here, which mostly report point estimates of Sharpe ratios from one test period. Our design uses them together: a small grid of four settings declared in advance and logged, a deflated Sharpe ratio, the Ledoit – Wolf test, and the same trading cost and timing rules for every strategy. The same discipline applies inside the model; the LSTM is trained on weekly holding -period returns charged for turnover at its weekly rebalancing date, so the cost it learns to avoid in training is the cost it pays in the backtest.

2.6 Theme 5- The 2022 shift: When bonds stopped protecting stocks

Most multi-asset portfolios, including 60/40 and any model trained on post-2000 data, rely on bonds rising when stocks fall. Brixton et al, (2023) show that this negative stock bond relationship is not a law of nature. Over the long run of history, it was often positive, and it turned negative around 2000,

during an era of low and stable inflation. When inflation returned in 2022, the relationship flipped back and both asset classes fell together.

This creates a Sharpe, testable question that deep-learning allocation studies we reviewed have not been able to answer, because their test periods end before 2022. For the 2022-fold, our LSTM is trained on 2007-2020 and validated on 2021, a period in which bonds mostly hedged stocks, and is then asked to allocate through 2022. Does a learned allocator cope with a regime it has never seen, or does it hold on to the lesson that bonds are safe? HRP re-estimates correlates every week from the past year, so it adjusts slowly but mechanically. Comparing the two through 2022, which we report as its own sub-period alongside 2015-2019,2019,2020, 2021 and 2023-2025, tells a practitioner something concrete about the limits of learning from history.

2.7 Synthesis: What is known, what is not, and our gap.

Table 2. How the closest studies were tested

Study	Asset	Test period	Includes 2022	HRP benchmark	Main limitation for an investor
Jiang et al,2017	Cryptocurrencies, 30-minute data	2016 -2017	No	No	Different market and time scale
Fischer& Krauss.,2018	S &P 500 stocks (stock picking)	1992- 2015	No	n/a	Profits faded after 2010 once costs counted
Zhang et al,2020	VTI, AGG, DBC, VIX index	2011- Apr 2020	No	No	VIX not investable; volatility overlay; short window
Uysal et al, 2024	ETFs	2017 - 2021	No	No (risk parity, 1/N)	Five-year window
This study	8 liquid multi-asset ETFd	2015 - 2025	Yes (reported separately)	Yes(primary)	Single asset universe (tested by dropping asset)

Four findings are well established. Estimation error dominates classical optimization. Simple rules are hard to beat. Structure-based rules such as HRP are more stable than optimizers. Neural networks can pick up real but small signals. What is not established is whether a learned allocator's advantage survive three practical tests at once: trading costs, a genuine regime change, and the possibility that it arose by chance. We therefore sharpen our research questions into three checks, that are simple to compute and easy for a decision maker to read:

1. Break-even cost (support RQ1). Instead of reporting results at a few cost levels only, we find the trading cost at which LSTM advantage over HRP disappears. If it sits near 10 basis points, it is fragile.
2. Cheap- adaptivity check (support RQ1 & RQ2). We add a 63- day inverse-volatility rule. If the LSTM cannot beat this, its extra complexity, is not earning its keep, because the same adaptivity is available almost for free.
3. Stress-window scorecard (support RQ3). We report each strategy's return and worst fall in three events chosen in advance: the Q4-2018 sell off, the COVID crash of February-March 2020, and the 2022 stocks- and-bonds fall. These sit alongside the five reporting sub-periods (2015-2019,2020,2021,2022 and 2023-2025), each reported separately.

Our contribution is therefore not a new architecture. It is a clean, cost-aware verdict on whether a published idea holds up in the market conditions investors now face, together with a clear account of

where it helps and where it fails. A negative result is just as useful in that it tells an investment committee that HRP, or even an inverse-volatility rule, is good enough.

Table 3. Research questions, the gap each addresses, and the planned test.

Question	Gap in the literature	Planned test
RQ1	DL allocators rarely compared with HRP or tested for significance	Net Sharpe Vs every benchmark; Ledoit-Wolf test; Deflated Sharpe; break-even cost; inverse-volatility check
RQ2	Unclear whether macro inputs help or just add noise	Rerun the LSTM on price-only features and compare
RQ3	Test windows end before 2022; single period results	Sub-periods 2015-2019,2020,2021,2022 and 2023-2025, each reported separately; costs 0-50 bps; stress-window scorecard; drop-one-asset runs

Conclusion of the review

The literature moves from optimizers that trust their inputs too much, to simple rules that trust them very little, to learning models that try to find a middle path. Each step has improved something, but each has also been tested on terms that suit it. The open question is not whether deep learning can allocate a portfolio; it clearly can. The question is whether it does so better than the best simple rule after costs, through a genuine regime change, and by more than one chance. Our study is built to answer exactly that, and to report the answer whichever way it falls.

3. Competitor Analysis

3.1 Who the competitors are;

For a research-track project, the competitors are the other ways an investor could solve the same problem. We grouped them into three kinds: classical rules (mean variance, 1/N,60/40, HRP) practical products and tools that package those rules (robo-advisers, risk-parity funds, and free libraries such as PYportfolioOpt; Martin,2021); and learning-based allocators from the research literature (Zhang et al,2020; Uysal et al, 2021); and deep reinforcement learning such as Jiang et al,2017).

3.2 Business case

Our intended users are small and medium-sized institutional investors such as pension schemes, insurers, endowments and cooperative or saving institution, including those in emerging markets such as East Africa that hold part of their offshore assets through global ETFs. Such institutions usually have small investment teams. They are increasingly offered “AI driven” allocation products, but they cannot easily judge whether the extra cost, complexity and governance burden is justified. A wrong choice is costly either way, but then paying for a model that adds nothing, or missing a real improvement in risk control.

The practical output of study is a simple adoption scorecard that such a committee can apply to any learned allocator, not only ours. Adopt only if all three conditions hold:

1. Net-of-cost Sharpe ratio above HRP's, with the difference statistically significant and deflated Sharpe Ratio above 0.95;
2. Break-even trading cost at least twice the cost the institution actually pays; and
3. No deeper loss than HRP in the pre-set stress windows, especially 2022.

This is where the market gap lies. Vendors publish attractive backtests; academic papers publish Sharpe ratio from one period. Few offers a transparent, repeatable test that a non-specialist committee can read. Our released code lets any institution rerun the scorecard on its own assets and costs.

3.3 SWOT analysis if competing approaches

Table 4 SWOT analysis of competing approaches

Competitor	Strength	Weakness	Opportunities	Threats
Mean-variance optimization (with shrinkage)	Clear theory; uses all information widely understood	Very sensitive to return estimates; concentrated weights; high turnover	Shrinkage and caps improve it	Simpler rules often match out of sample
1/N equal weight	No estimation; very low cost; hard to beat (DeMiguel et al, 2009)	Ignores risk; heavy in volatile assets	Honest, zero-skill baseline	Loses badly when one class crashes
60/40 stocks/bonds	Simple policy that boards understand; cheap	Fixed; relies on bonds hedging stocks	Commons reference for trustees	2022-type inflation shocks break the hedge
Volatility-managed/ risk parity products	Cheap adaptivity; mainstream institutional use	Gains less reliable out of sample (Cedeberg et al, 2020); some need leverage	Strong low-cost challenger to deep learning	Crowding; rising rate periods hurt bond-heavy mixes
End to end deep learning allocators (Zhang et al, 2020, Uysal et al, 2024)	Learn from many signals; optimize the real goal directly	Opaque; costly to validate; short tests; non-investable assets in some studies	Macro inputs hybrid designs with risk structure	Overfitting; results may fade out of sample, as with Fischer & Krauss, 2018
Deep reinforcement learning (Jiang et al, 2017)	Handles sequential decisions and cost naturally	Unstable training; many design choices; hard to audit	Growing research interest	Regulators and trustees wary of black boxes

1.3 Where our project fits:

Our project is not another product competing on performance. It sits between the simple rules and the learning models as an independent referee. Relative to HRP, it adds the possibility of learning from macroconditions and reacting faster. Relative to earlier deep learning allocators, it adds an investable universe, HRP and a cheap adaptive rule as benchmarks, a test period through 2025 in which 2020 and 2022 are reported as separate sub-periods, training that is consistent with weekly trading and its costs, formal significance tests, and break-even cost. Relative to reinforcement learning, it deliberately chooses a simpler, stable, auditable model. The unique value is the verdict and the scorecard, not the network.

2. Initial Code and Progress

3.1 The code is written in Python (Pandas, NumPy, scikit-learn, PyTorch) and supplied as separate files, ready to be posted to GitHub and shared with the professor. Every study setting lives in one file, config.py, so the settings in this report and in the code cannot drift apart. We wrote a short daily backtester of our own (backtest.py) so that the trade timing and cost rules are fully visible; vectorbt remains an optional cross-check. Following the review of our problem statement, the LSTM is now trained on the same weekly schedule on which it trades: one

sample per Friday decision, labelled with the return earned from that week's trade to the next, and charged for turnover at each weekly rebalancing date after prices have moved the previous weights. The Sharpe ratio in the loss is annualized over 52 week, so the turnover penalty tuned in training matches the costs paid in the backtest.

Table 5. Code files.

Config.py	All settings: assets, dates, 40% cap, 10bps cost, weekly schedule (52 periods a year), walk-forward years, the four-setting grid, sub-periods, stress windows
Data.py	Downloads and caches prices (Yahoo Finance) and macro series (FRED); lags macro data by one day
Features.py	48 price features (returns, volatility, momentum) plus 3 macro features
Benchmarks.py	HRP, 1/N, 60/40, minimum-variance, maximum-Sharpe (Ledoit-Wolf), 63-day inverse volatility
lstm_allocator	LSTM model with capped weights; weekly holding -period labels; Sharpe loss net of cost with weekly turnover; yearly walk-forward retraining with purge
Backtest.py	Daily simulation with drift and costs; metrics; break-even cost; Ledoit -Wolf test; Deflated Sharpe
Run.py	Runs everything and writes tables (including Table 6 and the sub-period tables) charts to outputs
Check.py	Mechanical checks: weights, 40% cap, trading timing, and training labels and turnover identical to the backtest

3.2 Status and initial results

The full pipeline (data, features, six benchmarks, LSTM walk forward, backtest, statistics and charts) has been updated for the weekly training design and run end to end again. As in the first cycle, the full run used a synthetic dataset, purely to check the machine before touching real data; those numbers are not findings and are not reported.

The checks (checks.py) confirmed that:

- Weights of every strategy sum to one on every decision date, and the LSTM and optimizers never exceed 40%.
- Trades occur the day after each decision, and costs are charged on actual turnover, including drift;
- The LSTMs weekly training label equals the return the backtest earns between two trades, and its training turnover equals the turnover the backtest charges, so the cost penalized in training is the cost paid in the backtest;
- The Ledoit- Wolf test, deflated Sharpe ratio, break-even cost, stress-window and sub-period tables (2015-2019, 2020, 2021, 2022 and 2023- 2025) all computed; and
- A full LSTM walk-forward with the complete grid (four settings, ten seeds) run on a standard two core CPU in about 20 minutes; training on weekly rather than daily samples has made it faster than in first cycle.

Table 6. Initial out-of-sample results, 2015 -2025, net of 10 pbs

Strategy	Annual returns	volatility	Sharpe	Max. drawdown	Turnover/year	Return,Jan-Oct 2022
HRP	0.0632	0.1069	0.6265	-0.2746	1.6727	0.0976
1/N	0.0199	0.1521	0.2055	-0.5220	1.1579	0.1393
60/40	0.0355	0.1950	0.2766	-0.5726	0.8089	0.0701
MinVar-LW	0.0826	0.0990	0.8509	-0.2022	1.8375	0.0839
MaxSharpe-LW	0.0501	0.1375	0.4244	-0.4129	8.2945	0.2111
InvVol-63d	0.0450	0.1256	0.4134	-0.3743	1.7171	0.1118
LSTM	0.0759	0.1200	0.6695	-0.3182	6.9013	0.0991

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